

A Single Layer Wideband U-Slot Microstrip Patch Antenna Array

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Abstract—The design and measured results of a 2×2 microstrip line fed U-slot rectangular antenna array are presented. The U-slot patches and the feeding network are placed on the same layer, resulting in a very simple structure. The advantage of the microstrip line fed U-slot patch is that it is easy to form the array. An impedance bandwidth (VSWR < 2) of 18% ranging from 5.65 GHz to 6.78 GHz is achieved. The radiation performance including radiation pattern, cross polarization, and gain is also satisfactory within this bandwidth. The measured peak gain of the array is 11.5 dBi. The agreement between simulated results and the measurement ones is good. The 2×2 array may be used as a module to form larger array.

Index Terms—Antenna arrays, single layer, U-slot, wide bandwidth.

I. INTRODUCTION

It is well known that microstrip antennas have very narrow impedance bandwidth, typically a few percent. Some wideband microstrip antenna arrays have been presented in [1]–[6] to overcome this drawback. In [1], an aperture coupled wideband microstrip antenna array is introduced. A wide-bandwidth patch antenna fed by an L-shaped probe is designed in [2]. A microstrip antenna array formed by microstrip line fed tooth-like-slot patches was proposed in [3]. One of the methods of widening the bandwidth is to cut a U-shaped slot on the patch. The coaxially fed U-slot rectangular patch antenna has been presented in [4]. The corresponding two element array has been proposed in [5]. The two patches are centre fed by probes. The patches are connected to microstrip lines through metallized holes. The other U-slot array is to use a U-slot patch proximity coupled by microstrip feed line terminated with a novel π -shaped stub [6]. Obviously, the above arrays must adopt multilayer structure. The multilayer structure will bring the difficulties in design and manufacture. As for the microstrip line fed structure, the advantage of it is that the patch and feeding network can be placed in the single layer, resulting in a simple structure. Yet the microstrip line fed U-slot antenna has not been reported.

In this letter, a microstrip line fed U-slot patch is proposed. A 2×2 array using this element is designed and manufactured. The measured results show an impedance bandwidth (VSWR < 2) of 18% ranging from 5.65 to 6.78 GHz.

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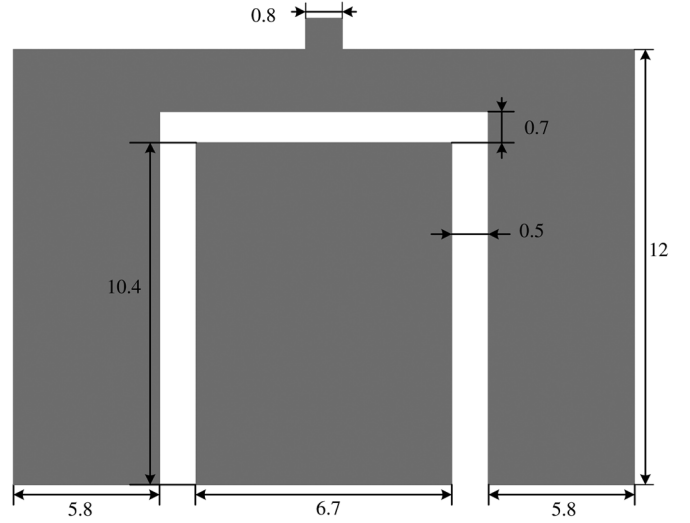


Fig. 1. The structure and detailed dimensions of a U-slot microstrip antenna element (Unit: mm).

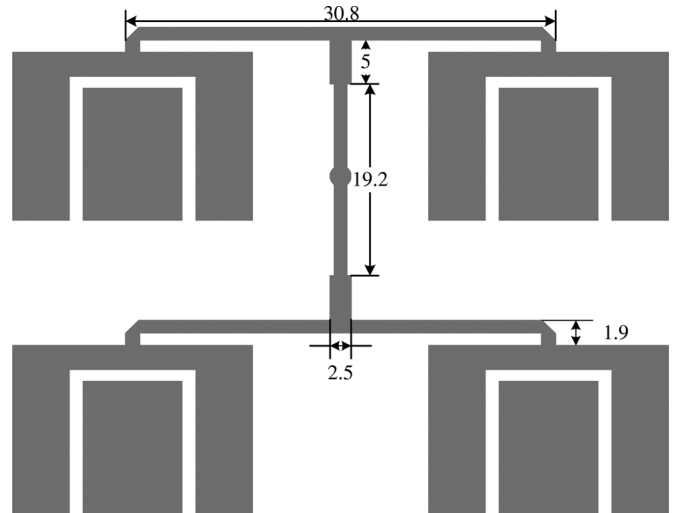


Fig. 2. The structure and detailed dimensions of the 2×2 U-slot microstrip array (Unit: mm).

II. DESCRIPTION OF ANTENNA ARRAY

The geometry and detailed dimensions of a single U-slot antenna element and 2×2 array are shown in Figs. 1–2. The dielectric substrate is with a relative permittivity ϵ_r of 2.7 and a thickness h of 3.5 mm. The spacings between two elements are $0.638\lambda_0$ along H plane and $0.621\lambda_0$ along E plane, respectively. The central frequency is 6.2 GHz. The design procedure of the antenna array can be summarized as follows:



Fig. 3. The photo of the 2×2 U-slot microstrip antenna array.

- 1) Decide the initial width and length of the patch according to the specified central frequency for a practical application. We suggest using the following approximate equation:

$$W = \frac{c}{2f_r \sqrt{\epsilon_e}} - 2\Delta w \quad (1)$$

$$L = \frac{c}{2f_r} \left(\frac{\epsilon_r + 1}{2} \right)^{-1/2} \quad (2)$$

where

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{10h}{W} \right)^{-1/2} \quad (3)$$

$$\Delta w = 0.412h \frac{(\epsilon_e + 0.3) \left(\frac{L}{h} + 0.264 \right)}{(\epsilon_e - 0.258) \left(\frac{L}{h} + 0.8 \right)} \quad (4)$$

f_r is the antenna operating frequency.

- 2) Design the slot. The U-slot is composed of two paralleled vertical rectangular slots and a horizontal rectangular slot. U-slot plays an important role to control the wideband behavior of the coupled patch antenna. There are three parameters to characterize the slots, namely slot length, slot position, and slot width. During the process of the optimization, we can exhibit a wider bandwidth.
- 3) Implement the optimization of the whole patch element using the GA (Genetic Algorithm) optimization function of the IE3D software [7]. GA was run for a total of 200 generations with a population size = 100 and a mutation rate = 0.1.
- 4) Initial design of the feed network.
- 5) Implementation of the integrated optimization to the whole array using the Genetic Algorithm (GA) optimization function of the IE3D software. GA was run for a total of 200 generations with a population size = 100 and a mutation rate = 0.1.

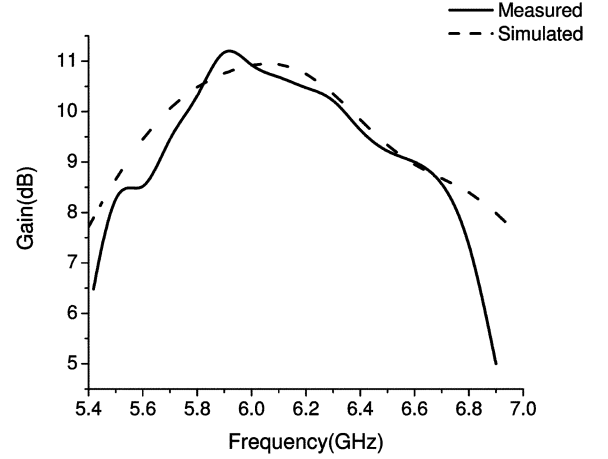


Fig. 4. The simulated and measured gain of 2×2 array.

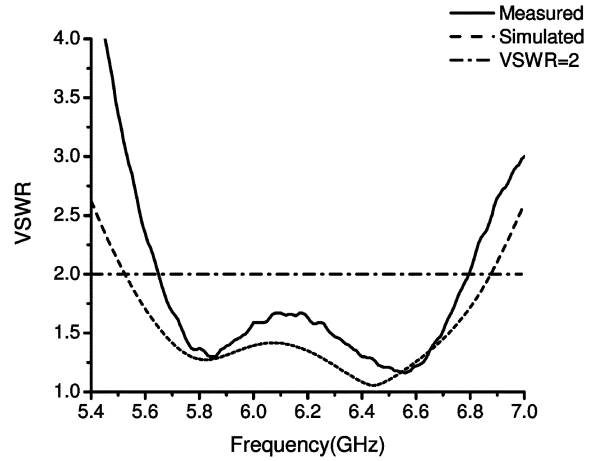


Fig. 5. The simulated and measured VSWR.

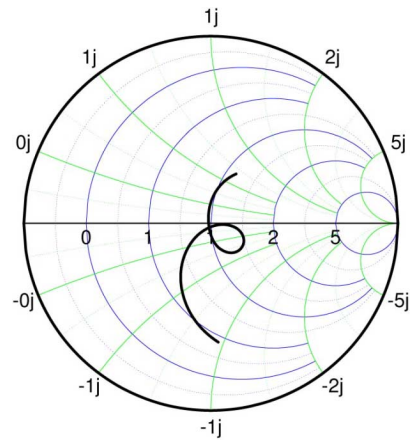


Fig. 6. The simulated result of Smith Chart.

III. SIMULATED AND MEASURED RESULTS

The photo of the array is shown in Fig. 3. Fig. 4 shows the simulated and measured gain curve of 2×2 array. **The measured peak gain of the array is 11.5 dBi.** The simulated and measured results of VSWR of this antenna element are shown in Fig. 5.

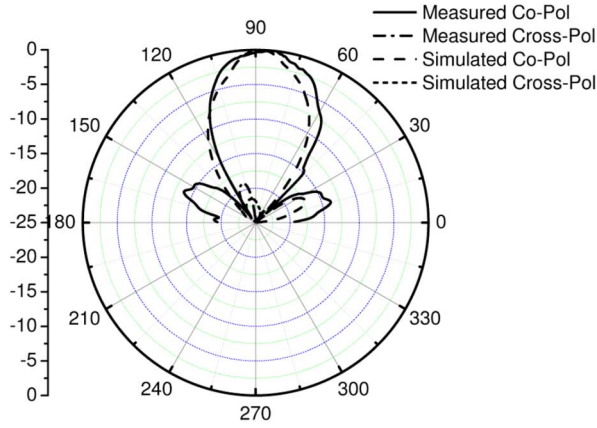


Fig. 7. The E-plane simulated and measured radiation pattern at central frequency.

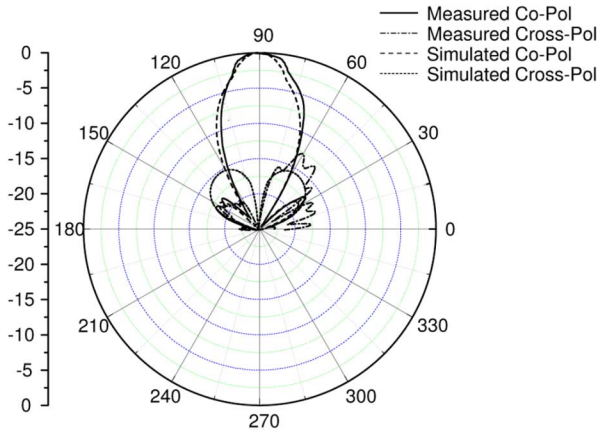


Fig. 8. The H-plane simulated and measured radiation pattern at central frequency.

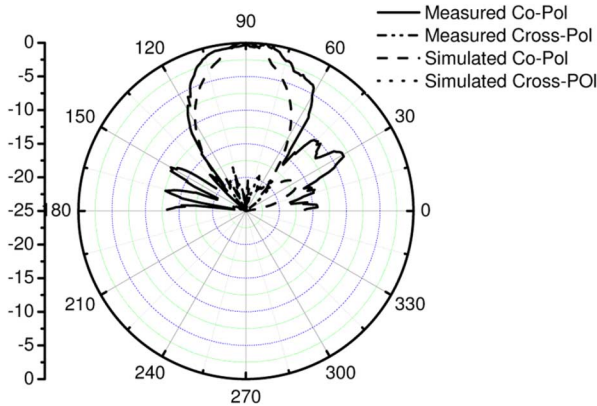


Fig. 9. The E-plane simulated and measured radiation pattern at 5.65 GHz.

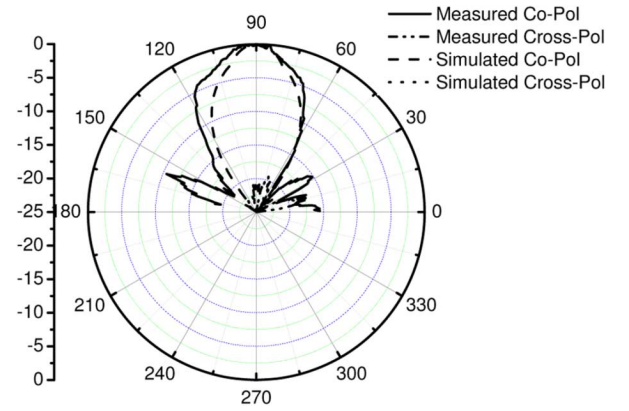


Fig. 10. The H-plane simulated and measured radiation pattern at 5.65 GHz.

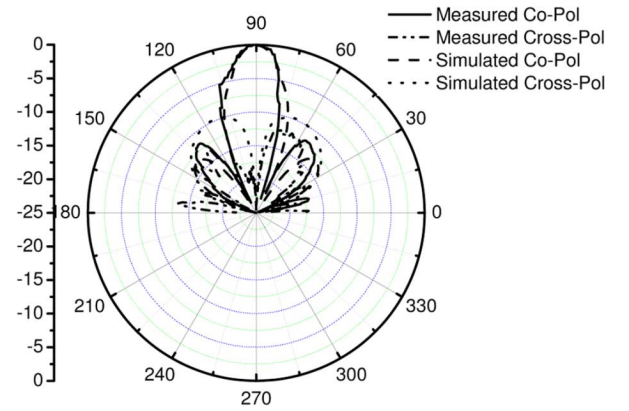


Fig. 11. The E-plane simulated and measured radiation pattern at 6.78 GHz.

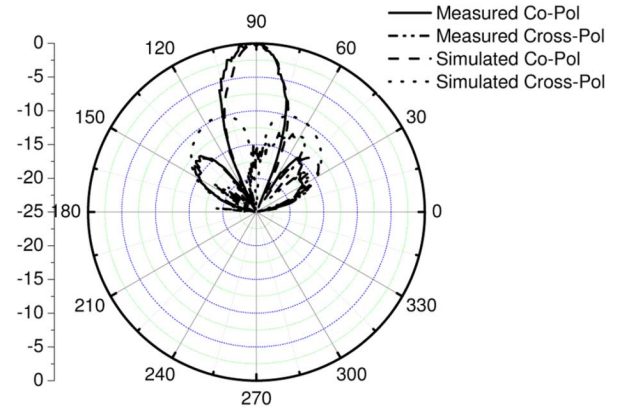


Fig. 12. The H-plane simulated and measured radiation pattern at 6.78 GHz.

The simulated result of Smith Chart is given in Fig. 6. Both figures show the performance of broad bandwidth. The bandwidth ($VSWR < 2$) is in the frequency range 5.65–6.78 GHz, corresponding to an impedance bandwidth of 18%. Figs. 7–8 show the radiation patterns of the antenna array at central frequency point. Figs. 9–10 show the radiation patterns of the antenna array at 5.65 GHz. Figs. 11–12 show the radiation patterns of the antenna array at 6.78 GHz. The radiation performance including

radiation pattern, cross polarization and gain is also satisfactory within this bandwidth.

IV. CONCLUSION

A 2×2 wideband microstrip antenna array using the microstrip line fed U-slot patch is presented. This array has the wide bandwidth and simple structure. This array can be easily expanded into a larger array through the array module method [8].

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